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THE USE OF VANADIUM AS A SCATTERING STANDARD FOR PULSED SOURCE NEUTRON SPECTROMETERS

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The Gaussian approximation for multiphonon cross-sections has been used in a calculation of the variation of vanadium cross-sections with incident neutron energy The results show that vanadium behaves as an elastic scatterer to within a few percent on pulsed neutron spectrometers with incident neutron energies up to 1 eV There is a calculated anisotropy in the scattering of 8% It is found that the scattering properties of vanadium at 77 K and 293 K differ by a maximum of 1% except for neutron energies <15 meV

1. Introduction

In neutron scattering experiments, vanadium is often used as a calibration sample. It is assumed that vanadium behaves as an incoherent elastic scatterer which therefore scatters isotropically. The incoherent cross-section of vanadium is well known [1] and the results of a vanadium run can be used to determine the absolute cross-section of the scattering atoms [2]. The vanadium run can also be used to calibrate the incident neutron intensity and the detector efficiencies [3]. The method is particularly useful because the vanadium run can be done with the same geometry as the sample run, thereby eliminating any necessity of calculating geometric corrections.

However, it is not clear that treating vanadium as an isotropic elastic scatterer is a good approximation, particularly for neutrons of greater than thermal energies. The elastic scattering in a vanadium sample at room temperature, for incident neutrons of 1 eV, is less than 10% of the total scattering cross-section (fig. 1). Pulsed neutron sources produce intense fluxes of high energy 'epithermal' neutrons and it seems worthwhile to investigate the errors which are introduced by the assumption that vanadium behaves as an ideal elastic isotropic scatterer.

The Gaussian approximation [4] for multiphonon inelastic scattering has been used in an evaluation of the scattering properties of vanadium. Total, differential and partial differential cross-sections of vanadium are calculated in section 3. A general expression for the count rate in a pulsed neutron instrument is derived in section 4. In section 5, the behaviour of a hypothetical ideal isotropic elastic vanadium calibration sample is compared with that calculated in the Gaussian approximation for three general classes of pulsed neutron instruments: direct geometry, inverse geometry and total scattering.

2. The Gaussian approximation

The Gaussian approximation for multiphonon cross-sections has been discussed in detail by Sjolander [4]. The treatment given by Marshall and Lovesey [5] is followed here. The total cross-section for incoherent scattering is

$$\frac{d^2\sigma}{d\Omega dE} = \frac{\sigma_{inc}}{4\pi}\frac{K_1}{K_0}\frac{N}{2\pi h}\exp\langle U^2\rangle\int_{-\infty}^{\infty}\exp\langle UV_0\rangle\exp(-i\omega t)dt \quad (1)$$

where K_0 = wavevector of incident neutrons, K_1 = wavevector of scattered neutrons, K = scattering vector, $U = -i\,K \cdot U_0(0)$, $V_0 = i\,K \cdot U_0(t)$; $\exp\langle U^2\rangle$ is the Debey-Waller factor, $U_0(t)$ is the displacement of an atom from its mean position at time t. $\langle\rangle$ denotes thermal averaging.

The multiphonon terms are derived by expanding $\exp\langle UV_0\rangle$

$$\exp\langle UV_0\rangle = \sum_{n=0}^{\infty} \langle UV_0\rangle^n/n!. \tag{2}$$

The $n = 0$ term is responsible for elastic scattering, $n = 1$ for one phonon scattering etc. For a cubic crystal such as vanadium,

$$\langle UV_0\rangle = \langle \boldsymbol{K}\cdot \boldsymbol{U}_0(t)\boldsymbol{K}\cdot \boldsymbol{U}_0(0)\rangle = (\hbar K^2/2M)\gamma(t), \tag{3}$$

where

$$\gamma(t) = \int_{-\infty}^{\infty} \mathrm{d}\omega \frac{Z(\omega)}{\omega} n(\omega) \exp(-\mathrm{i}\omega t). \tag{4}$$

$Z(\omega)$ is the phonon density of states.

$$n(\omega) = [\exp(\hbar\omega/KT) - 1]^{-1}, \tag{5}$$

M is the mass of the vanadium atom.
For small values of t

$$\exp(-\mathrm{i}\omega t) \approx 1 - \mathrm{i}\omega t + \tfrac{1}{2}(\mathrm{i}\omega t)^2 \quad \text{and} \quad \gamma(t) \approx \gamma(0) + \mathrm{i}t - \frac{2t^2\varepsilon}{3\hbar}, \tag{6}$$

where

$$\varepsilon = \tfrac{3}{2}\int_0^{\infty} \mathrm{d}\omega Z(\omega)\hbar\omega\left[n(\omega) + \tfrac{1}{2}\right] \tag{7}$$

is the mean kinetic energy per atom.
Thus,

$$\langle UV_0\rangle^n \approx \left(\frac{\hbar K^2\gamma(0)}{2M}\right)^n \left(1 + \frac{\mathrm{i}t}{\gamma(0)} - \frac{2t^2\varepsilon}{3\hbar\gamma(0)}\right)^n \tag{8}$$

$$\approx \left(\frac{\hbar K^2\gamma(0)}{2M}\right)^n \exp\left[n\left(\frac{\mathrm{i}t}{\gamma(0)} - \tfrac{1}{2}\Delta^2 t^2\right)\right], \tag{9}$$

where

$$\Delta^2 = \frac{4\varepsilon}{3\hbar\gamma(0)} - \frac{1}{\gamma^2(0)}. \tag{10}$$

The partial differential scattering cross-section is given by substituting eqs. (9) and (10) in eq. (1) and performing the integration over t. The n phonon contribution to the partial differential cross section is:

$$\left(\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega \mathrm{d}E}\right)_{\text{incoh}}^{n\text{ phonon}} = \frac{N\sigma_\mathrm{i}}{4\pi\hbar}\frac{K_1}{K_0}\exp\left(-\frac{\hbar K^2}{2M}\gamma(0)\right)\frac{1}{\Delta}\times\frac{1}{n!}\left(\frac{\hbar K^2\gamma(0)}{2M}\right)^n \frac{1}{(2\pi n)^{1/2}}$$
$$\times \exp\left(-\frac{\omega^2}{2\Delta^2 n} - \frac{n}{2\Delta^2\gamma^2(0)} + \frac{\omega}{\gamma(0)\Delta^2}\right). \tag{11}$$

The multiphonon terms are represented by a Gaussian function of ω with mean values and widths depending on n. In the present work, this Gaussian approximation was used for $n \geqslant 2$. The one-phonon terms are calculated exactly using the experimental values for $Z(\omega)$ given by Glaser et al. [6]. The cross-sections calculated should therefore be accurate at low neutron energies, where one-phonon processes

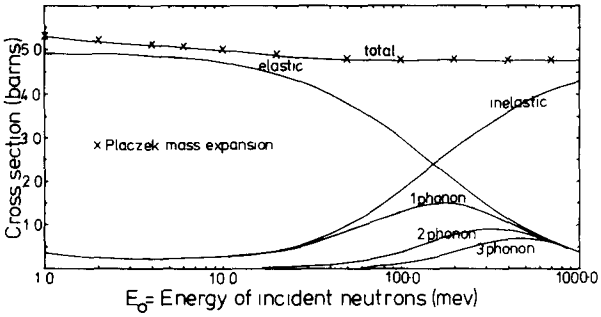


Fig 1 Total scattering cross-section at 293 K

dominate the inelastic scattering. The Gaussian approximation is valid at large values of ω, i.e. for large incident energy. As is shown in the following section, where comparison is possible there is good agreement with cross-sections calculated by other methods.

3. Calculation of vanadium cross-sections

The parameters [7] $\sigma_i = 4.98$ b, $M = 50.942$ amu, $\theta_D = 390$ K were used in all calculations. The Debey–Waller factor for vanadium was calculated using the Debye approximation for $Z(\omega)$. This gave satisfactory agreement with the experimental values given by Kamal et al. [8].

Figs. 1 and 2 show the total scattering cross-sections calculated at 293 K and 77 K as a function of incident neutron energy E_0. It was necessary to include $n \leqslant 20$ phonon processes at $T = 293$ K although $n \leqslant 10$ was sufficient for $T = 77$ K. The various contributions to the cross-section are also illustrated. The total cross-sections agree closely with those calculated using the Placzek mass expansion technique [5]. At high values of E_0 the total cross-section becomes equal to the free atom value of $\sigma_i/(1 + m/M)^2 = 4.79$ b.

The partial differential cross-sections for incident neutron energy $E_0 = 1000$ meV are shown in fig. 3. The free atom recoil values calculated from eq. (2)

$$\frac{\Delta E}{E_0} = 1 - \frac{1}{(1+A)^2}\left[\cos\phi + \left(A^2 - \sin^2\phi\right)^{1/2}\right]^2, \tag{12}$$

where $A = M/m = 51$, are also drawn in for comparison. The calculated energy loss of the neutrons is smaller than the free atom recoil value and the inelastic line is broadened. The broadening increases with scattering angle.

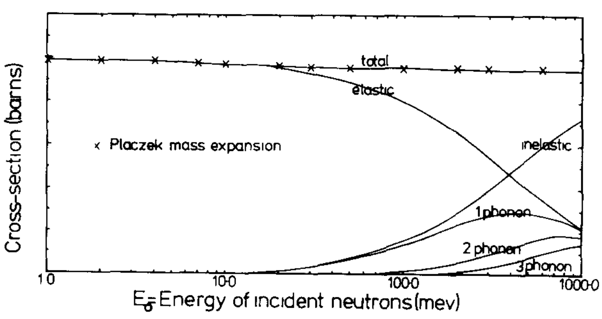


Fig 2 Total scattering cross-section at 77 K

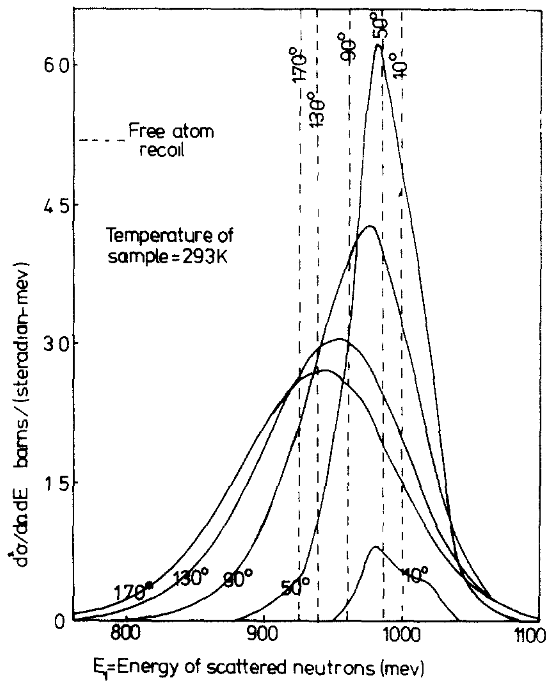


Fig 3. Inelastic contribution to the partial differential cross-section for neutrons of 1000 meV incident energy

Figs. 4 and 5 show the differential cross-sections calculated as a function of incident neutron energy for scattering angles of 10°, 50°, 90°, 130° and 170°. These were calculated by integrating $d^2\sigma/d\Omega dE$ over all scattered energies. Day et al. [10] have given an analytic expression for $d\sigma/d\Omega$ which is valid for $E_0 > kT$. This was derived by Egelstaff from the paper by Placzek [9].

$$\frac{4\pi}{\sigma}\frac{d\sigma}{d\Omega} = 1 - \frac{\alpha}{\varepsilon}\left(1 - \frac{1}{2A} - \frac{3\alpha}{8\varepsilon}\right) + \frac{\overline{K}}{3\varepsilon A}. \tag{13}$$

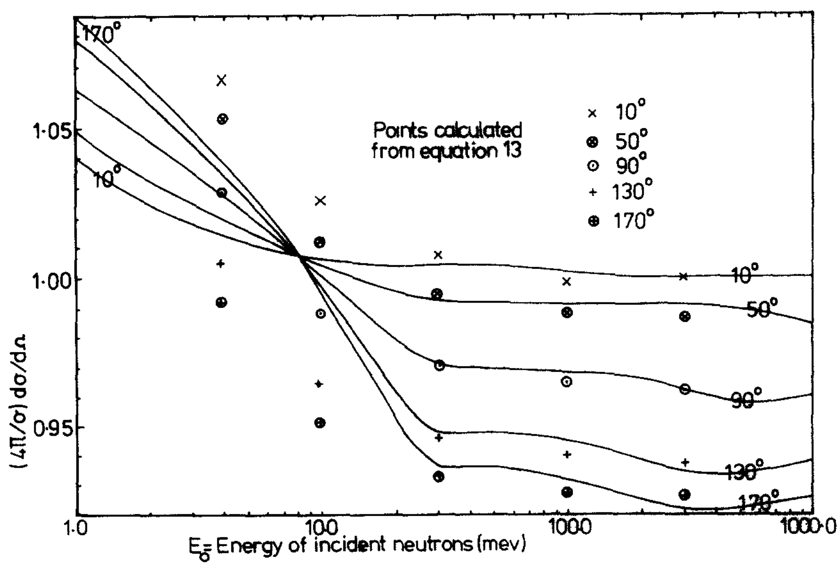


Fig 4 Differential cross-section at 293 K

where $\bar{K} = 1.63$ at $T = 293$ K, $\bar{K} = 3.02$ at $T = 77$ K, $A = 51$, $\alpha/\epsilon = (2/A)(1 - \cos\phi)$, $\varepsilon = E_0/kT$ and ϕ is the scattering angle.

Points calculated from this expression are also shown in figs. 4 and 5 There is good agreement for $E_0 > 30$ meV but at lower energies eq. (13) predicts a behaviour of $d\sigma/d\Omega$ both qualitatively and quantitatively different from the exact values calculated. The corresponding increase in total cross-section (fig. 1) has been observed by Granada et al. [11]. The low energy behaviour of $d\sigma/d\Omega$ is caused by phonon annihilation processes in vanadium. As $E_0 \to 0$, the inelastic cross-section increases because of the $1/K_0$ term in eq. (11). The inelastic component also increases with scattering angle giving the anisotropy illustrated. This anisotropic rise in cross-section below 15 meV can be almost entirely eliminated if the vanadium is cooled to 77 K (fig. 5). However, it is noticeable that for $E_0 > 15$ meV the calculated cross-sections differ by a maximum of ~ 1% at 77 K and 293 K

4. Count rates in pulsed neutron experiments

A schematic diagram of a pulsed neutron instrument is shown in fig. 6. Subscripts 0 refer to incident neutrons and 1 to scattered neutrons. The neutron time-of-flight from source to detector is t and the detector subtends a solid angle $d\Omega$ about the scattering angle ϕ. The time of flight (TOF) is

$$t = L_0/V_0 + L_1/V_1. \tag{14}$$

Thus

$$V_1 = \frac{L_1 V_0}{t V_0 - L_0}. \tag{15}$$

The scattering vector K is given by

$$K^2 = K_0^2 + K_1^2 - 2K_0 K_1 \cos\phi, \tag{16}$$

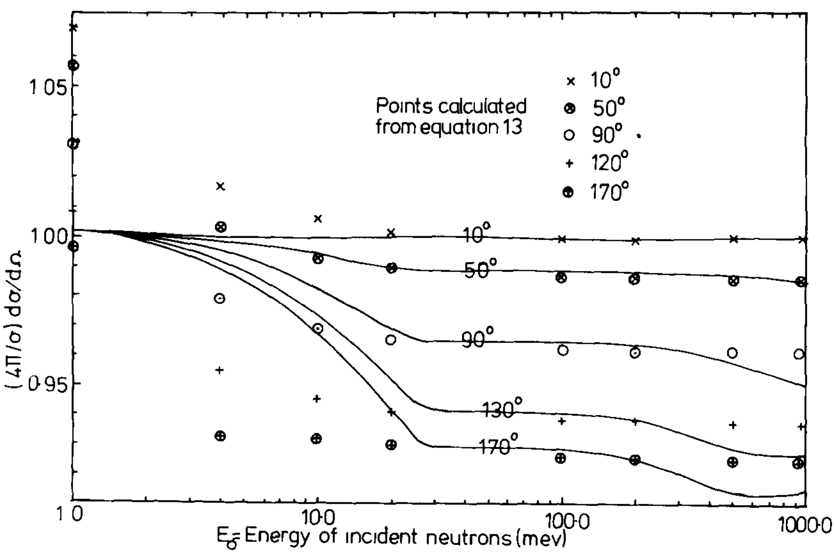


Fig 5 Differential cross-section at 77 K

and the energy lost by the neutron is

$$\hbar\omega = \frac{\hbar^2}{2m}\left(K_0^2 - K_1^2\right).$$

K and ω can be written as functions of t and V_0, for a given value of ϕ.

$$K(t,V_0) = \frac{mV_0}{\hbar}\left(1 + \frac{L_1^2}{(tV_0 - L_0)^2} - \frac{2L_1\cos\phi}{tV_0 - L_0}\right)^{1/2}, \tag{17}$$

$$\omega(t,V_0) = \frac{mV_0^2}{2\hbar}\left(1 - \frac{L_1^2}{(tV_0 - L_0)^2}\right). \tag{18}$$

The partial differential scattering cross-section per atom of the sample is $\mathrm{d}^2\sigma(K,\omega)/\mathrm{d}\Omega\mathrm{d}\omega$. The probability that a neutron with incident velocity V_0 will be scattered into the detector and have a TOF between t and $t+\Delta t$ is

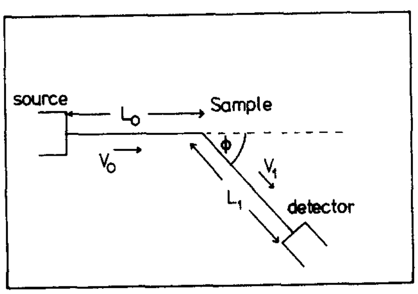
$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\mathrm{d}\omega}\frac{\mathrm{d}\omega}{\mathrm{d}t}\Delta t\Delta\Omega = \frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\mathrm{d}\omega}(V_0,t)\frac{mV_0^3}{\hbar}\frac{L_1^2}{\left(tV_0 - L_0^3\right)}\Delta t\Delta\Omega, \tag{19}$$

[from eqs. (17) and (18)].

Let the probability that the neutron is detected be $\eta(V_1)$ and let the flux of neutrons incident on the sample with velocities between V_0 and $V_0 + \mathrm{d}V_0$ be $I(V_0)\mathrm{d}V_0$. Then the rate at which neutrons are detected in a time channel of width Δt at t is

$$D(t)\Delta t = \Delta t\int_{L_0/t}^{\infty}\mathrm{d}V_0 I(V_0)\frac{mV_0^3}{\hbar}\frac{L_1^2}{(tV_0 - L_0)^3} \times\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\mathrm{d}\omega}(V_0,t)\,\eta\left(\frac{L_1V_0}{tV_0 - L_0}\right) \tag{20}$$

This derivation neglects attenuation and multiple scattering in the sample.



V_0 = velocity of incident neutrons
V_1 = velocity of scattered neutrons
L_0 = flight path from source to sample
L_1 = flight path from sample to detector
ϕ = scattering angle
$\Delta\Omega$ = solid angle subtended by detector

Fig. 6 Schematic diagram of an instrument on a pulsed neutron source

5. Use of vanadium as a calibration sample

We now consider the errors which are introduced by the assumption that vanadium scatters elastically and isotropically. If the scattering is purely elastic,

$$\frac{d^2\sigma}{d\Omega d\omega}(V_0,t)=\frac{d\sigma}{d\Omega}(V_0)\delta(\omega)$$
$$=\frac{d\sigma}{d\Omega}(V_0)\frac{\hbar(tV_0-L_0)}{mV_0^2 t}\left[\delta\left(\frac{L_1+L_0}{t}-V_0\right)+\delta\left(\frac{L_1+L_0}{t}+V_0\right)\right]. \quad (21)$$

Substituting eq. (21) in to eq. (20) gives

$$D(t)\Delta t=I\left(\frac{L}{t}\right)\frac{d\sigma}{d\Omega}\left(\frac{L}{t}\right)\eta\left(\frac{L}{T}\right)\left(\frac{L}{t^2}\right)\Delta t, \quad (22)$$

where $L=L_0+L_1$.

Thus the count rate is proportional to the product of the incident intensity and detection efficiency appropriate for neutrons of velocity L/t. When inelastic scattering occurs, this simple relationship breaks down. It is worth considering the effects on three different types of pulsed source instrument.

5.1. Direct geometry instruments

In a direct geometry inelastic spectrometer the energy of the incident neutrons is defined by means of a monochromator or a chopper. The incident flux may be written

$$I(V_0)=\delta(V_0-V_m)I(V_0), \quad (23)$$

where V_m is the velocity selected. From eq. (20)

$$D(t)\Delta t=I(V_m)\frac{mV_m^3}{\hbar}\frac{L_1^2}{(tV_m-L_0)^3}\frac{d^2\sigma}{d\Omega d\omega}(V_m,t)\eta\left(\frac{L_1V_m}{tV_m-L_0}\right)\Delta t. \quad (24)$$

The apparent energy of the detected neutrons is

$$E=\tfrac{1}{2}m(L/t)^2. \quad (25)$$

This is the energy scattered neutrons appear to have, if it is assumed that all scattering is elastic. E is related to the energies of the incident and scattered neutrons by the equation

$$\frac{1}{E^{1/2}}=\frac{1}{(1+\alpha)}\left(\frac{1}{E_0^{1/2}}+\frac{\alpha}{E_1^{1/2}}\right), \quad (26)$$

where $\alpha=L_1/L_0$ is the ratio of the scattered and incident flight paths. The apparent broadening of the elastic line in the TOF spectrum depends upon the actual energy broadening of the scattered neutrons (i.e. the distribution of E_1 values about E_0) and upon α. As $\alpha \to 0$ the apparent width of the elastic line $\to 0$.

A second effect of inelastic scattering is caused by the dependence of the detector efficiency on the energies of scattered neutrons. This effect is most important at high values of E_0 where most of the scattering is inelastic. In the worst case calculated (fig. 3), a scattering angle of 170° and $E_0=1000$ meV, the average energy of the scattered neutrons is reduced to ~ 940 meV. Examination of detector efficiency curves [12] shows that this corresponds to a change in detector efficiency of ~ 4% for a wide range of detectors. Thus if it is assumed that the detection efficiency is $\eta(E)$ instead of the correct value $\eta(E_1)$, the error introducted is less than 4% even in this extreme case. At forward scattering angles or lower values of E_0, the error will be even smaller.

5.2. Inverse geometry instruments

In an inverse geometry inelastic spectrometer the scattered neutrons are energy selected. The detector efficiency may be described by

$$\eta(V_A)=\delta(V_A-V_1)\eta(V_1), \tag{27}$$

where V_A is the velocity selected by the analyser.

$$\eta(V_A)=\frac{tV_0-L_0}{L_1-V_At}\delta\left(V_0-\frac{L_0V_A}{V_At-L_1}\right)\eta(V_A). \tag{28}$$

Substituting into eq. (20) gives

$$D(t)\Delta t=\frac{m}{\hbar}\frac{V_A^3L_0}{(tV_A-L_1)^2}\frac{\mathrm{d}^2\sigma}{\mathrm{d}\Omega\mathrm{d}\omega}(t,V_A)I\left(\frac{L_0V_A}{tV_A-L_1}\right)\eta(V_A). \tag{29}$$

Only neutrons with incident energies $E_0=E_1=E$ would be detected if the scattering was elastic.

Again we consider the worst case: $E_0=1000$ meV, $E_1=940$ meV, $\phi=170°$. From eq. (26), with $\alpha=1/8$, $E=993$. Thus if it is assumed that the incident spectrum is that corresponding to E, this differs from the true incident spectrum by

$$\Delta F=F(E)-F(E_0).$$

For epithermal neutrons where

$$F(E)\propto 1/E,$$

the fractional error introduced is

$$\Delta F/F=0.007.$$

A value of $E_1=940$ meV for the scattered energy selected is much larger than is typical. A more usual value is $E_1\approx 100$ meV. Calculation shows that for $E_0=100$ meV, phonon creation and phonon annihilation processes make comparable contributions to inelastic scattering at $T=293$ K. The net result is that the mean energy of the scattered neutrons is very close to that of the incident neutrons. Consequently the error introduced by the assumption that scattering is elastic should be totally negligible for $E_1\approx 100$ meV. Even in the largely hypothetical worst case of large E_1 and large ϕ, the error is less than 1%

5.3. Total scattering

In this case neither the incident neutrons nor the scattered neutrons are energy selected. As previously mentioned, elastic scattering gives a count rate proportional to the product of the incident spectrum and the detector efficiency. If inelastic scattering is present, the observed TOF spectrum is changed both in the total number of counts summed over all channels and in the distribution of these counts among the different time channels. The way in which the counts are distributed over the time channels depends upon the value of α. If $\alpha\ll 1$, then $E\approx E_0$ and the distribution of neutrons among the time channels depends essentially on the incident spectrum (neglecting effects caused by varying detector efficiency).

If $\alpha\gg 1$ time dispersion of inelastically scattered neutrons could lead to a significant distortion of the elastic TOF spectrum. In general $\alpha\ll 1$ for total scattering instruments.

Eq. (20) was numerically integrated to give the count rate as a function of t. The incident spectrum was taken from Hobbis et al. [13] and the detection efficiency used was that appropriate to a 40 atm 0.762 cm diameter ^{3}He detector [12]. The count rate was also calculated for an ideal isotropic elastic vanadium scatterer [eq. (22)]. The ratio of the two count rates was plotted as a function of $E(=\frac{1}{2}mL^2/t^2)$.

The values $L_0=8$ m, $L_1=1$ m, $\alpha=1/8$ corresponding to a typical geometry were used.

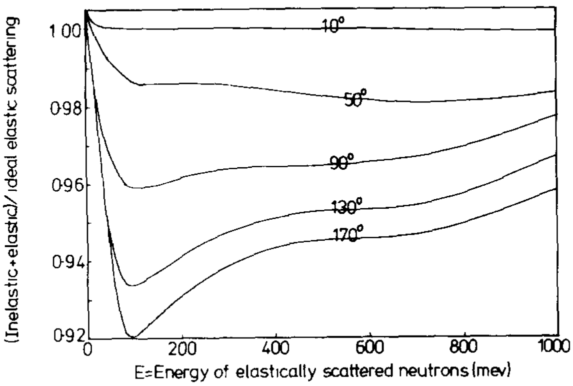


Fig 7 Total scattering at 293 K

Figs 7 and 8 show the results obtained at 293 K and 77 K. It is again noticeable that there is only a minor difference in the results when the vanadium is cooled to liquid nitrogen temperature except for $E < 15$ meV. The rise in the ratio at high values of E and ϕ is due to down scattering of neutrons. At high values of E, inelastically scattered neutrons will, on average, have a lower value for E_1 than elastically scattered neutrons and a correspondingly higher detection efficiency. The dominant overall feature of figs. 7 and 8 is the anisotropy of the scattering.

6. Discussion

It can be seen from the results in sections 3 and 5 that vanadium behaves as an elastic scatterer to a very good approximation in all three types of pulsed source spectrometer. The worst case considered showed a variation of ~ 5% from elastic behaviour and in most cases the discrepancies are within 2%. However the scattering is significantly anisotropic, varying typically by 8% over the range of scattering angles from 0° to 180°.

These results pertain to an ideal situation in which neutrons undergo a single scattering event with no attenuation in the sample. Multiple scattering increases as the absorption cross-section decreases [14,15] and will therefore be more serious at large values of E_0. Standard corrections can be applied for attenuation effects and although multiple scattering is difficult to correct for, two general comments can be made. The apparently elastic behaviour of the scattering should be unchanged by multiple scattering. Multiple scattering generally reduces any anisotropy of scattering and should therefore make vanadium approximate more closely to an isotropic scatter.

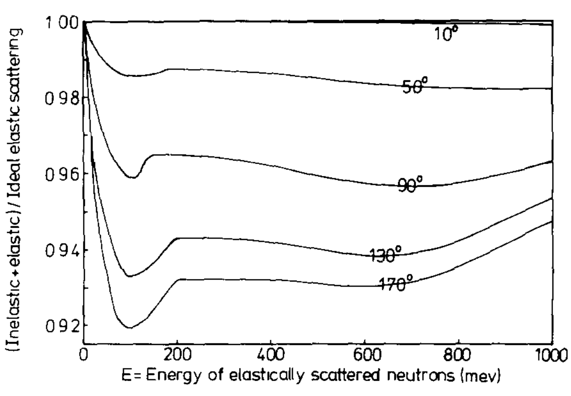


Fig 8 Total scattering at 77 K

A final perhaps surprising conclusion, independent of multiple scattering effects, is that cooling vanadium from 293 K to 77 K has no significant effect on its scattering properties on pulsed source instruments, except for incident neutron energies below 15 meV.

I would like to thank W.G. Williams for suggesting this problem and for his critical reading of the manuscript.

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